

When Standard Retrieval Optimizations Fail: A Cost-Aware Ablation Study of RAG Pipelines for Financial Document Question Answering

*Anote AI Research Fellowship 2026
Supervised by Natan Vidra, CEO, Anote AI*

ABSTRACT

We present a seven-condition controlled ablation study of standard RAG retrieval improvements on FinanceBench, evaluated across 133 questions from 10-K, 10-Q, and earnings transcript filings. Conditions include semantic chunking, recursive chunking, cross-encoder reranking, metadata enrichment, HyDE query expansion, and a full hybrid. The central finding is negative: no condition improves substantially over a fixed 512-token chunking baseline on ROUGE-L F1. The hybrid (C6, F1 = 0.179) exceeds the baseline (C0, F1 = 0.177) by 0.002 at 4.0x the latency; the F1 spread across all seven conditions is 0.012. We show this failure is structural: fixed-token splitting destroys table row integrity in SEC filings and no retrieval technique recovers passages that were never coherently chunked. A secondary finding corrects an evaluation methodology problem: ROUGE-L understates model performance on numeric questions by 10 to 16 times. Exact Match reveals semantic chunking achieves 34.0% correct numerical answers on metrics-generated questions, the highest of any condition, despite having the lowest ROUGE-L (0.172). All results are single-run estimates; differences this small require multi-seed replication to confirm statistical significance.

Keywords: retrieval-augmented generation, financial document QA, chunking strategies, reranking, FinanceBench, SEC filings, negative results.

1. INTRODUCTION

Enterprise question answering over SEC filings is a commercially important problem that standard RAG pipelines handle poorly. Setty et al. [1] show that GPT-4-Turbo with a standard retrieval system incorrectly answers or refuses to answer 81% of questions on FinanceBench even when the answer is present in the document. Our C0 baseline, GPT-4o with fixed 512-token chunking and cosine similarity retrieval, achieves mean ROUGE-L F1 = 0.177 across 133 questions and 0.028 on metrics-generated numeric questions specifically.

The natural response is to deploy standard RAG improvements: better chunking, reranking, metadata, query expansion, and hybrid combinations. This paper tests whether that response works on financial documents. It largely does not, and understanding why is the contribution.

Four contributions. First, a seven-condition controlled ablation with each technique varied independently plus a hybrid. Second, a negative result: F1 spread across all conditions is 0.012; the hybrid improves 0.002 over baseline at 4x latency; the failure is structural. Third, a positive finding within the negative: Exact Match reveals semantic chunking achieves 34.0% correct numerical answers, the highest of any condition, despite the lowest ROUGE-L F1 (0.172); ROUGE-L understates numeric performance by 10 to 16 times. Fourth, differentiated practical recommendations by metric.

2. RELATED WORK

2.1 Financial Document Question Answering

FinanceBench [1] provides 10,231 questions over SEC filings; GPT-4-Turbo fails 81% under standard retrieval. FinQA [4] established multi-step numerical reasoning difficulty over financial tables. FinAgentBench [5] treats each table as a single chunk unit, directly motivated by the structural problem identified here. Dyer et al. [10] show 19% accuracy with ada-002 embeddings alone on FinanceBench.

2.2 Retrieval Optimizations for Financial Documents

Setty et al. [2] test chunking, query expansion, metadata, and reranking on FinanceBench, the direct predecessor. Our work adds a fully controlled single-variable ablation, a hybrid condition, and a negative result. Patel et al. [3] find BM25 outperforms dense retrieval and query expansion provides limited benefit for precise numeric queries, consistent with our C5 result. Banerjee et al. [7] focus on metadata-driven retrieval. Kim et al. [8] demonstrate multi-reranker ensembling.

2.3 Metric Selection

ROUGE-L is the standard FinanceBench metric. Section 5.2 shows it understates model capability on numeric questions by 10 to 16 times relative to Exact Match. Prior work does not note this discrepancy.

3. METHODOLOGY

3.1 Dataset

We evaluate on the publicly available FinanceBench split [1], 150 questions spanning metrics-generated (47; multi-step numeric), novel-generated (40; extractive and interpretive), and domain-relevant (46; qualitative analysis). We use 133 of 150 questions after excluding 17 whose source PDFs were not retrievable: Adobe 10-Ks (2015, 2016, 2017, 2022); Johnson and Johnson (2022 10-K, Q4 2022 and Q2 2023 earnings, 2023 8-K); and MGM Resorts (Q4 2022 earnings). This exclusion may introduce selection bias.

3.2 Ablation Design

Table 1 lists the seven conditions. Each of C1 through C5 varies exactly one component from C0. C6 combines all five techniques.

Table 1. Ablation conditions. Each of C1-C5 varies exactly one component from C0.

Condition	Chunking	Reranking	Metadata	Query Exp.
C0 Baseline	Fixed 512-token	None	None	None
C1 Semantic	Semantic boundary	None	None	None
C2 Recursive	Recursive 2k-token	None	None	None
C3 +Reranking	Fixed 512-token	BGE-v2-m3	None	None
C4 +Metadata	Fixed 512-token	None	Prefix	None
C5 +Query Exp.	Fixed 512-token	None	None	HyDE
C6 Hybrid	Recursive 2k-token	BGE-v2-m3	Prefix	HyDE

3.3 Pipeline Components

C0 Fixed Chunking.

TokenTextSplitter, chunk_size = 512, chunk_overlap = 50. Baseline for all comparisons.

C1 Semantic Chunking.

SemanticChunker with all-MiniLM-L6-v2 at the 95th-percentile breakpoint threshold, producing variable-length chunks at semantic boundaries.

C2 Recursive Chunking.

RecursiveCharacterTextSplitter with separator hierarchy (section headers, paragraphs, lines, sentences, words, characters) and chunk_size = 2,000.

C3 Reranking.

k = 20 base retrieval; BAAI/bge-reranker-v2-m3 as cross-encoder; retains top 5 for generation.

C4 Metadata Enrichment.

Company ticker, fiscal year, and filing type prepended to chunks and queries as a structured prefix.

C5 HyDE Query Expansion.

GPT-4o generates a hypothetical analyst answer per question; embedded and merged with the original query result set.

C6 Hybrid.

Recursive chunking plus metadata prefix plus BM25 and dense EnsembleRetriever (50/50) plus cross-encoder reranking plus HyDE.

3.4 Evaluation Metrics

ROUGE-L F1 (standard FinanceBench metric; primary accuracy signal), Exact Match (gold answer appears as substring of prediction; added because Section 5.2 shows ROUGE-L understates numeric performance), and mean wall-clock latency in seconds. Cost per query and confidence intervals will be added following multi-seed replication.

4. RESULTS**4.1 Baseline**

C0 achieves ROUGE-L F1 = 0.177, Exact Match = 0.128, latency = 12.3 s across 133 questions. Table 2 disaggregates by question type.

Table 2. C0 Baseline by question type. Exact Match on metrics-generated questions (0.277) substantially exceeds ROUGE-L F1 (0.028); see Section 5.2.

Question Type	N	ROUGE-L F1	Exact Match	Note
metrics-generated	47	0.028	0.277	13/47 correct; near-zero ROUGE-L
novel-generated	40	0.261	0.050	Prose retrieval adequate
domain-relevant	46	0.257	0.022	Qualitative questions retrieved adequately
All	133	0.177	0.128	10x F1 gap between question types

4.2 Full Ablation

Table 3 presents all seven conditions. The F1 range is 0.012 (0.167 to 0.179). All results are single-run; differences this small should be treated as directional until replicated across multiple seeds.

Table 3. Complete ablation results. Single-run estimates. Delta F1 relative to C0.

Condition	ROUGE-L F1	Exact Match	Latency (s)	Delta F1
C0 Baseline	0.177	0.128	12.3	ref.
C1 Semantic	0.172	0.135	37.0	-0.005
C2 Recursive	0.177	0.120	11.2	0.000
C3 +Reranking	0.175	0.128	35.3	-0.002
C4 +Metadata	0.170	0.105	10.8	-0.007
C5 +Query Exp.	0.167	0.098	13.2	-0.010
C6 Hybrid	0.179	0.098	49.8	+0.002

4.3 Results by Question Type

Table 4. ROUGE-L F1 and metrics-generated Exact Match by condition. The EM row shows the metric discrepancy on numeric questions.

Metric	C0	C1	C2	C3	C4	C5	C6
metrics F1	0.028	0.021	0.026	0.024	0.024	0.017	0.018
metrics EM	0.277	0.340	0.277	0.319	0.128	0.098	0.234
novel F1	0.261	0.250	0.257	0.258	0.269	0.254	0.260
domain F1	0.257	0.259	0.263	0.259	0.232	0.245	0.273

The C1 Exact Match on metrics-generated questions (0.340; 16 of 47 correct) is the most important result in Table 4. Semantic chunking achieves the highest correct numerical answer rate of any condition despite the lowest overall ROUGE-L F1 (0.172). This inversion is analyzed in Section 5.2.

4.4 Cost and Latency

C0 (12.3 s) and C2 (11.2 s) are the fastest. C1 (37.0 s) and C3 (35.3 s) are comparable. C6 (49.8 s) is the slowest. On ROUGE-L F1, C0 and C2 are the only non-dominated conditions. On Exact Match, C1 (EM = 0.135) is the only enhanced condition that exceeds C0 (EM = 0.128), at 3.0x the latency.

5. ANALYSIS

5.1 Structural Chunking Failure Is Technique-Independent

Financial tables in SEC filings have column headers at the top, row labels at the left, and numeric values in right-aligned columns, often spanning multiple pages. When PDF text is split at fixed token boundaries, the three semantic components of a table cell (column header, row label, and value) are routinely placed in separate chunks. A similarity search for 'capital expenditure FY2018' retrieves the chunk with the row label but not the chunk with the value. The answer does not exist as a retrievable unit.

No cross-encoder reranker can surface a value never attached to its label. No HyDE expansion matches a chunk of decontextualized numeric values. No metadata prefix locates a passage that does not exist. The C6 hybrid, combining all five techniques, achieves metrics-generated ROUGE-L F1 = 0.018, below the C0 baseline (0.028). C2 recursive chunking uses 2,000-token chunks and achieves metrics-generated F1 = 0.026, essentially the same as C0. SEC filing tables frequently span more than 2,000 tokens when linearized. The fix is not larger chunks; it is table-aware parsing that preserves row-column structure independently of text splitting.

This has a direct practical implication: if a financial RAG system fails on questions like 'What was capital expenditure in FY2022?', adding reranking or query expansion will not fix it. The root cause is that cash flow statement values are not reaching the model as coherent, contextually complete units. That requires parsing tables as structured objects with preserved row-column associations.

5.2 Semantic Chunking Achieves the Highest Numeric Accuracy; ROUGE-L Hides It

C1 achieves the lowest overall ROUGE-L F1 (0.172) yet the highest Exact Match on metrics-generated questions (0.340; 16 of 47 correct). Under C0, metrics-generated EM is 0.277 at ROUGE-L F1 = 0.028. Semantic chunking more reliably retrieves complete table rows, since it splits on semantic boundaries rather than arbitrary token counts. The model computes correct answers. But GPT-4o's step-by-step explanations suppress ROUGE-L overlap with short gold answers.

The mechanism: a response calculating '24.26' in 60 words achieves Exact Match = 1.0 but ROUGE-L F1 approximately 0.04. Semantic chunking produces this pattern more often because semantic boundaries more frequently

keep the row label and its value together in a single chunk. ROUGE-L understates numeric performance by 10 to 16 times across all conditions (Exact Match 0.234 to 0.340 against ROUGE-L F1 0.017 to 0.028).

A practitioner selecting conditions by ROUGE-L alone would reject semantic chunking (0.172, lowest F1) and select the hybrid (0.179, highest F1), arriving at the opposite of the correct recommendation for numeric accuracy. Financial QA benchmarks should report Exact Match alongside ROUGE-L, disaggregated by question type.

5.3 Practical Recommendations

On the ROUGE-L and latency plane, C0 and C2 are the only non-dominated conditions. On the Exact Match and latency plane, C1 is non-dominated relative to C0. Avoid C5 regardless of metric: it decreases accuracy on all measures while adding cost. C4 similarly decreases both accuracy metrics. Practical summary: for ROUGE-L or cost-constrained settings, use C0 or C2. For numeric Exact Match, use C1. For all settings, table-aware document parsing is the intervention with the highest expected marginal return and is not represented in any condition tested here.

6. ABLATION DISCUSSION

6.1 A1: Semantic vs. Fixed Chunking (C1 vs. C0)

ROUGE-L F1 = 0.172 (-0.005); Exact Match = 0.135 (+0.007); latency = 37.0 s. The only condition improving any accuracy metric over C0. Metrics-generated EM = 0.340, the highest of any condition. The ROUGE-L decrease reflects that semantic chunks are longer and semantically broader than gold answer spans, suppressing LCS overlap while preserving correct numerical values.

6.2 A2: Recursive vs. Fixed Chunking (C2 vs. C0)

ROUGE-L F1 = 0.177 (identical to C0); Exact Match = 0.120 (-0.008); latency = 11.2 s. A 4x increase in chunk size does not recover financial table values. Structural failure is invariant to chunk size within the range tested. C2 is a safe swap for C0.

6.3 A3: Reranking (C3 vs. C0)

ROUGE-L F1 = 0.175 (-0.002); Exact Match = 0.128 (unchanged); latency = 35.3 s (+23 s). Metrics-generated EM = 0.319, second-highest. The reranker surfaces correct chunks more reliably when coherent units exist but cannot address structural incompleteness.

6.4 A4: Metadata (C4 vs. C0)

ROUGE-L F1 = 0.170 (-0.007); Exact Match = 0.105 (-0.023); latency = 10.8 s. Both accuracy metrics decrease. The structured prefix appears to shift query embeddings away from content-specific terms toward structural metadata tokens.

6.5 A5: Query Expansion (C5 vs. C0)

ROUGE-L F1 = 0.167 (-0.010, largest single-condition decrease); Exact Match = 0.098 (-0.030); latency = 13.2 s. Metrics-generated F1 = 0.017, lowest of any condition. Consistent with Patel et al. [3]. GPT-4o's hypothetical answer likely contains plausible but incorrect figures that retrieve numerically similar but wrong passages.

6.6 A6: Hybrid (C6 vs. C0)

ROUGE-L F1 = 0.179 (+0.002, the only positive delta); Exact Match = 0.098 (-0.030); latency = 49.8 s (4.0x C0). Metrics-generated EM = 0.234, below C0 (0.277). The ROUGE-L gain reflects domain-relevant improvement (0.273 vs. 0.257)

from recursive chunking and metadata. The Exact Match decrease reflects HyDE degrading numeric retrieval. C6 is the best ROUGE-L condition and the worst Exact Match condition.

7. LIMITATIONS

We enumerate limitations explicitly.

1. Single-run estimates throughout. All F1 and EM figures are single-run. The ROUGE-L spread across all conditions (0.012) may lie within single-run variance. Differences between conditions should be treated as directional until replicated.
2. 17 questions excluded. Company pages that block automated PDF retrieval may differ systematically in document structure, introducing selection bias of unknown direction.
3. General-purpose embedding model. all-MiniLM-L6-v2 was not trained on financial text. Domain-specific embeddings may change relative condition orderings.
4. Cost per query not yet reported. OpenAI API usage logs are available but not yet aggregated per condition.
5. Table-aware parsing not tested. This paper identifies it as the necessary fix but does not implement or measure it.
6. Single generation model. All conditions use GPT-4o.
7. Hardware-specific latency. Figures measured on Windows with Anaconda Python 3.11.

8. FUTURE WORK

Table-aware chunking is the highest-priority next experiment. Treating each financial table as a structured object with preserved row-column associations, following FinAgentBench [5], would directly address the structural failure in Section 5.1. We also plan to test whether semantic chunking (C1) complements table-aware parsing, since C1's strong Exact Match performance suggests it already captures more complete table context than fixed-token splitting.

Additional planned work: three-seed replication to establish variance estimates; cost-per-query reporting from aggregated API logs; evaluation of financial-domain embedding models; question-type routing to direct numeric questions to a table-extraction pipeline and qualitative questions to the current RAG pipeline; and extension to the full 10,231-question FinanceBench benchmark.

9. CONCLUSION

We present a seven-condition controlled ablation of RAG retrieval techniques on FinanceBench. The ROUGE-L F1 range across all conditions is 0.012, from 0.167 to 0.179, with no condition achieving a practically meaningful improvement over the 0.177 baseline. The failure is structural: fixed-token text splitting destroys table row integrity in SEC filings, and no combination of retrieval techniques recovers values that were never coherently chunked.

The most actionable finding is the semantic chunking inversion: lowest ROUGE-L F1 (0.172) but highest metrics-generated Exact Match (0.340; 16 of 47 numeric questions correct). ROUGE-L understates model capability on numeric questions by 10 to 16 times. A practitioner selecting retrieval conditions by ROUGE-L alone would reject the best condition for numeric accuracy.

Practical recommendations: use semantic chunking (C1) when numeric Exact Match is the primary concern; use C0 or C2 when latency or cost is constrained; avoid query expansion (C5) and metadata enrichment (C4); and invest in table-aware document parsing before any further retrieval optimization.

References

- [1] Setty, V., et al. (2023). FinanceBench: A New Benchmark for Financial Question Answering. arXiv:2311.11944.
- [2] Setty, V., Vidra, N., et al. (2024). Improving Retrieval for RAG-based QA on Financial Documents. arXiv:2404.07221.
- [3] Patel, A., et al. (2026). From BM25 to Corrective RAG. arXiv:2604.01733.

- [4] Chen, Z., et al. (2021). FinQA: Numerical Reasoning over Financial Data. EMNLP 2021.
- [5] Kim, S., et al. (2025). FinAgentBench. ACM ICAIF 2025.
- [6] Pal, A., et al. (2025). FinTagging. arXiv:2505.20650.
- [7] Banerjee, S., et al. (2025). Metadata-Driven RAG for Financial QA. arXiv:2510.24402.
- [8] Kim, J., et al. (2025). Multi-Reranker for FinanceRAG. arXiv:2411.16732.
- [9] FinGEAR. (2025). EMNLP Findings 2025.
- [10] Dyer, E., et al. (2024). Finance is a Jungle. arXiv:2411.07142.